Def: **Retraction** : the streamline of a mfld.

A retraction on a mfld $\mathcal{M}$ is a smooth map $R : T\mathcal{M} \to \mathcal{M}$

$(x, v) \mapsto R_x(v)$

such that each curve $c(t) = R_x(tv)$ satisfies $c(0) = x$ and $c'(0) = v$.

Equivalent definition: Retraction R is a smooth map $R: T\mathcal{M} \to \mathcal{M}$ s.t.

$R_x(0) = x$ and $DR_x(0) = Id_{T_x\mathcal{M}}$.

$R_x : T_x\mathcal{M} \to \mathcal{M}$

$DR_x : T_0T_x\mathcal{M} \to \mathcal{M}$, $T_0T_x\mathcal{M} \cong T_x\mathcal{M}$

Pf: former $\Leftrightarrow$ latter : $c'(0) = R_x'(tv) = DR_x|_0(v) = v \quad \forall v$

$\Leftrightarrow DR_x = Id_{T_x\mathcal{M}}$

prop : Since DR_x is of full rank, by Inverse function theorem, R_x is a local diffeo at the origin.

It is obvious that the exponential map is a Retraction However, to get the exponential map, we have to solve the geodesic ODE: $\ddot{\gamma}^k + \underset{\text{||}}{\Gamma_{ij}^k} \dot{\gamma}^i \dot{\gamma}^j = 0 \quad \forall k$ which will cost

$$\nabla_{\dot{\gamma}} \dot{\gamma} = 0$$

a lot. So we do a relaxation of the condition and only demand R to be smooth. So $c(t)$ is a smooth map.

First order optimization algorithm

4.1 Taylor expansion on curves

$c(0)=x,\ c'(0)=v$ $\qquad g'(t)=df(c'(t))=\langle \nabla f, c'(t)\rangle_g$

$c: I \to M,\ f: M \to \mathbb{R}$ smooth $\Rightarrow g = f\circ c : I \to \mathbb{R}$ smooth.

with Taylor expansion $g(t) = g(0) + tg'(0) + O(t^2)$

$$\Rightarrow f(c(t)) = f(x) + t\langle \nabla f(x), v\rangle_x + O(t^2)$$

If $c(t) = R_x(tv)$, $f(R_x(s)) = f(x) + \langle \nabla f(x), s\rangle_x + O(\|s\|_x^2)$

4.2 First order optimality conditions

Def: $x \in M$ is critical or stationary for a smooth function $f: M \to \mathbb{R}$ if $(f\circ c)'(0) \geq 0$ for all smooth curves c on M s.t. $c(0)=x$ $\Leftrightarrow$ $df|_x = 0$

This def is the same as in $\mathbb{R}^n$

prop 4.5 Any local minimizer of a smooth function $f: M \to \mathbb{R}$ is a critical pt of f.

pf: Assume $(f\circ c)'(0) < 0 \Rightarrow f(c(t)) = f(c(0)) + \int_0^t (f\circ c)'\tau\, d\tau < f(x)$ $\forall t$ $\Rightarrow$ contradict to minimizer

prop 4.6 $f: M \to \mathbb{R}$ smooth. then x = critical pt $\Leftrightarrow \nabla f(x) = 0$

one line proof: $(f\circ c)'(0) = Df(x)[v] = \langle \text{grad} f(x), v\rangle_x$

4.3 Riemannian gradient descent.

Recall: gradient descent in $\mathbb{R}^n$: $f: \mathbb{R}^n \to \mathbb{R}$. To minimize f, Let $x_{k+1} = x_k - \alpha_k \nabla f(x_k)$

where α_k is called step length. $k = 0, 1, 2 \cdots$

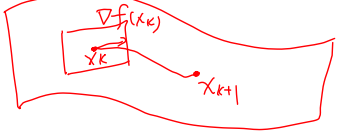
Def: Riemannian gradient descent (RGD)

we have to choose a retraction otherwise it is meaningless to talk about next pt with only direction given

given $x_0 \in M$ and a retraction R on M, iterate $x_{k+1} = R_{x_k}(-\alpha_k \nabla f(x_k))$, $k = 0, 1, \ldots$

To determine the coefficient α_k, we use Line search:

$g(t) = f(R_{x_k}(-t\nabla f(x_k)))$ where $t \mapsto R_{x_k}(-t\nabla f(x_k))$ is a curve on M



Line search Common Method:

① Fixed step-size $\alpha_k = \alpha \quad \forall k$

- advantage : easy
- drawbacks : α large $\Rightarrow$ not converge; α small $\Rightarrow$ converge slowly

② Optimal setp-size $\alpha_k = \underset{t>0}{\operatorname{argmin}}\, g(t)$

- advan : valid
- draw : expensive

③ Backtracking : guess $t_0 > 0$, if not proper, $t_1 = \tau t_0$, $t_2 = \tau^2 t_0$ .... until acceptable.

and let $\alpha_k = t_i$

Below is the ordinary steplength line search in $\mathbb{R}^n$

Method	Main requirement	Evaluation cost or limitation
Fixed safe step	Curvature and direction bounds	No trials, but possibly conservative steps
Exact search	Attained scalar minimum	Cheap for special scalar problems
Backtracking	Function-value decrease test	Function evaluations at rejected trials
Weak Wolfe	Decrease and derivative tests	Trial derivatives and possible step expansion

All four approaches can give sufficient decrease. Constants and costs differ.

Basic assumptions of RGD

[A]: $\exists\, f_{low} \in \mathbb{R}$ s.t $f \geq f_{low}$, $\forall x \in M$ — f has minimum

[B]: At each iteration, the algorithm achieves sufficient decrease for f. That is, $\exists$ a constant $c > 0$ s.t, $\forall k$, $f(x_k) - f(x_{k+1}) \geq c \| \nabla f(x_k) \|^2$ — algorithm satisfies Sufficiat decreas

The Line search must ensure the assumption holds and it can be done under some conditions on f and the retraction (Later)

When [A, B] hold, RGD produces pts with small grad for any initialization $x_0 \in M$

prop 4.7 : f satisfies [A], $x_0, x_1 \cdots x_n \cdots$ satisfying [B] with constant c

$\Rightarrow \lim_{k\to\infty} \| \operatorname{grad} f(x_k) \| = 0 \Rightarrow$ all accumulation pts (if exists) (→ may not exist) are critical pts.

Furthemore, $\forall K \geq 1$, $\exists\, k$ in $0, 1, \cdots K-1$ s.t $\| \nabla f(x_k) \| \leq \sqrt{\frac{f(x_0) - f_{low}}{c}} \frac{1}{\sqrt{K}}$

pf: almost the same as in $\mathbb{R}^n$. $f(x_0) - f_{low} \geq f(x_0) - f(x_K)$

$$= \sum_{k=0}^{K-1} f(x_k) - f(x_{k+1}) \geq Kc \min_{K=0\cdots K-1} \|\nabla f(x_K)\|^2$$

Since we have $f(x_{k+1}) \leq f(x_k)$, $\Rightarrow f(x_0) - f_{low} \geq \sum_{k=0}^{\infty} f(x_k) - f(x_{k+1})$ and RHS is nonnegative each term $\Rightarrow 0 = \lim_{k\to\infty} f(x_k) - f(x_{k+1}) \geq c \lim_{K\to\infty} \|\nabla f(x_K)\|^2$

→ the manifold may not be compact!

Now assume x is an accumulation pt $\Rightarrow \exists$ subseq of $\{x_k\}$, called $\{x_{n_k}\}$ s.t $x_{n_k} \to x \Rightarrow \|\nabla f(x)\| = 0 \Rightarrow$ accumulation pts are critical.

Regularity conditions and iteration complexity: Find the condition of f and R to realize B

By Taylor expansion along a curve, $f(x_{k+1}) = f(R_{x_k}(s_k)) = f(x_k) + \langle \nabla f(x_k), s_k \rangle + O(\|s_k\|^2)$

in RGD, $s_k = -\alpha_k \nabla f(x_k)$

$$f(x_{k+1}) - f(x_k) = -\alpha_k \|\nabla f(x_k)\|^2 + O(\alpha_k^2 \|\nabla f(x_k)\|^2)$$

if $O(\|s_k^2\|)$ stays under control, we deduce a guarantee of B

↓

look at assumption C

C given $S \subset TM$. $\exists L > 0$ s.t for all $(x,s) \in S$, we have

$$f(R_x(s)) \leq f(x) + \langle \nabla f(x), s \rangle + \frac{L}{2}\|s\|^2$$

f has the same property as L-smooth in $\mathbb{R}^n$

Recall in $\mathbb{R}^n$, $\|\nabla f(x) - \nabla f(y)\| \leq L\|x-y\|$, we call f is L-smooth. $\Rightarrow f(x+s) \leq f(x) + \nabla f(x)^T s + \frac{L}{2}\|s\|^2$

on Manifold, "$x+s$" is not defined so we replace it by $R_x(s)$ and Replace $\langle \cdot,\cdot \rangle_R$ by $\langle \cdot,\cdot \rangle_g$.

On summary the three assumptions are

- f has lower bound A
- algorithm satisfies SD B
- f, R is "L-smooth" C

① prop 4.8: If f, R satisfying C on a set $S \subset TM$ with constant L.

② If $(x_0,s_0), (x_1,s_1), (x_2,s_2) \cdots$ generated by RGD all lie in S

→ all satisfying C

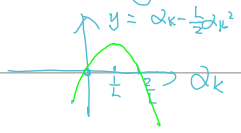
(2) with step-sizes $\alpha_k \in [\alpha_{min}, \alpha_{max}] \subset (0, 2/L)$, then [B] holds with

$$c = \min\left(\alpha_{min} - \frac{L}{2}\alpha_{min}^2 \,,\ \alpha_{max} - \frac{L}{2}\alpha_{max}^2\right) > 0$$

In particular, for $\alpha_k = \frac{1}{L}$ constant, we have $c = 1/2L$

pf: L smooth means $f(x_{k+1}) \le f(x_k) + \langle \nabla f(x_k), s_k \rangle + \frac{L}{2}\|s_k\|^2$

replace s_k by $-\alpha_k \nabla f(x_k)$, we get $f(x_k) - f(x_{k+1}) \ge (\alpha_k - \frac{L}{2}\alpha_k^2)\|\nabla f(x_k)\|^2$



Set $\alpha_k \in (0, 2/L)\ \forall k$, we have $\alpha_k - \frac{L}{2}\alpha_k^2 \ge \min\left(\alpha_{min} - \frac{L}{2}\alpha_{min}^2 \,,\ \alpha_{max} - \frac{L}{2}\alpha_{max}^2\right) > 0$

→ prop 4.7 — has lower bound — L smooth

Coro 4.9 Let f satisfies [A], f o R satisfies [C] with constant L on S,

$(x_0, s_0), (x_1, s_1) \cdots$ generated by RGD lie in S with $\alpha_k \equiv 1/L$, then

$$\lim_{k \to \infty} \|\nabla f(x_k)\| = 0$$

furthermore, for all $K \ge 1$, there exists k in $0, \cdots K-1$ s.t

$$\|\nabla f(x_k)\| \le \sqrt{2L(f(x_0) - f_{low})}\,\frac{1}{\sqrt{K}}$$

pf: Directly from Prop 4.7

Def 4.10 A sublevel set of f is a set $\{x \in M : f(x) \le \alpha\}$ for some α.

4.5 Backtracking line search

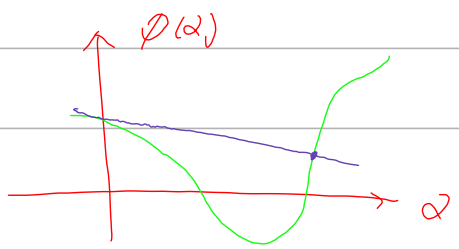
Recall: Armijo Sufficient Decrease in $\mathbb{R}^n$

For line search, $\phi(\alpha) = f(x_k + \alpha p_k)$, where p_k is the descent direction

Choose $c_1 \in (0,1)$ and require $\phi(\alpha) \le \phi(0) + c_1\alpha\phi'(0)$

↓ when $\nabla f(x_k) = g$, $p = -g$

$f(x - \alpha g) \le f(x) - \alpha c_1 \|g\|^2$



Satisfies ASD

Now back to M, we similarly have ASD to be $f(x) - f(R_x(-\alpha \nabla f(x))) \geq r\alpha \|\nabla f(x)\|^2$ since R gives us the streamline on mfld.

Backtracking algorithm : parameters $\tau, r \in (0,1)$. The same as $\mathbb{R}^n$.

Input $x \in M$, $\bar{\alpha} > 0$

Set $\alpha = \bar{\alpha}$

while ASD is not satisfied

Set $\alpha = \tau \alpha$

Output α

Lemma 4.12 Let f be smooth function on M, let C hold for $f \circ R$ with L on $\{(x, -\alpha \nabla f(x)) : \alpha \in [0, \bar{\alpha}]\}$ where $x \in M$ and initial step size is $\bar{\alpha} > 0$. Then Backtracking with parameters $\tau, r \in (0,1)$ outputs a step-size α s.t

$$f(x) - f(R_x(-\alpha \nabla f(x))) \geq r \min\left(\bar{\alpha}, \frac{2\tau(1-r)}{L}\right) \|\nabla f(x)\|^2$$

after computing at most $\max\left(1, \left\lfloor 2 + \log_\tau \left(\frac{\bar{\alpha} L}{2(1-\alpha)}\right)\right\rfloor\right)$ times of retractions and cost function evaluations.

The Euclidean version is below when p is the descent direction and $c_1 = r$ in this problem

Smoothness gives

$$f(x + \alpha p) \leq f(x) + \alpha g^T p + \frac{L\alpha^2}{2} \|p\|^2.$$

Therefore Armijo holds whenever

$$\boxed{0 < \alpha \leq A := \frac{2(1 - c_1)(-g^T p)}{L \|p\|^2}.}$$

Since $A > 0$, repeated multiplication by ρ eventually reaches this interval. For $p = -g$ and $c_1 = 1/2$, $A = 1/L$. The number of rejected trials is at most

$$\max\left\{0, \left\lceil \frac{\log(\bar{\alpha} L)}{\log(1/\rho)} \right\rceil\right\}.$$

pf: The same as the form in $\mathbb{R}^n$. Use L-smoothness combining ASD

Corollary 4.13 f is A, for R is C, every iteration uses Backtracking with initial step sizes $\bar{\alpha}_0, \bar{\alpha}_1, \bar{\alpha}_2 \cdots$ and $\{(x_k, -\alpha \nabla f(x_k)) \mid \alpha \in [0, \bar{\alpha}_k]\}$ is in S.

If $\liminf_{k\to\infty} \bar{\alpha}_k > 0$, then $\lim_{k\to\infty} \|\nabla f(x_k)\| = 0$

Furthermore, for all $K \geq 1$, there $\exists\ k \in \{0, 1, \cdots K-1\}$ s.t

$$\|\nabla f(x_k)\| \leq \sqrt{\frac{f(x_0) - f_{low}}{r \min(\bar{\alpha}_0, \cdots \bar{\alpha}_{k-1}, \frac{2\tau(1-r)}{L})}} \frac{1}{\sqrt{K}}$$

pf: Backtracking guarantees decrease in the form

$$f(x_k) - f(x_{k+1}) \geq c_k \|\nabla f(x_k)\|^2 \quad \text{with} \quad c_k = r \min\left(\bar{\alpha}_k, \frac{2\tau(1-r)}{L}\right)$$

Use the same trick in 4.7 we have $f(x_0) - f_{low} \geq \sum_{k=0}^{\infty} c_k \|\nabla f(x_k)\|^2$

$\Rightarrow \lim_{k\to\infty} c_k \|\nabla f(x_k)\|^2 = 0$. Since $\liminf_{k\to\infty} c_k > 0$ only could have $\lim_{k\to\infty} \|\nabla f(x_k)\|^2 = 0$

4.6 local convergence

Def a method enjoys global convergence if it generates sequences that converge regardless of their initialization x_0.

Def a method is local convergence if the initial pt x_0 is close to some solution / critical pt x_*, then the sequence $\{x_k\}$ will converge to x_*.

In practice, RGD is global convergence (not proved yet!)

Below, we will just prove one thing:

Thm: RGD with constant step size is local convergence.

(a.l.l)
Def 4.14 (at least linear convergence) Consider metric space with distance dist, a sequence $a_0, a_1, a_2 \cdots$ converges at least linearly to a^* if there $\exists$ positive reals $\varepsilon_0, \varepsilon_1, \varepsilon_2 \cdots$ converging to zero s.t. $\text{dist}(a_k, a^*) \le \varepsilon_k$ and $\lim_{k\to\infty} \frac{\varepsilon_{k+1}}{\varepsilon_k} = \mu \in (0,1)$

change the sequence $\{\varepsilon_k\}$, we get diffaen't μ's. The infimum over such μ is the linear convergence factor. If the latter is zero, call the convergence superlinear.

(a.l.q)
Def 4.15 (at least quadratic convergence) $\lim_{k\to\infty} \frac{\varepsilon_{k+1}}{\varepsilon_k^2} = \mu$.

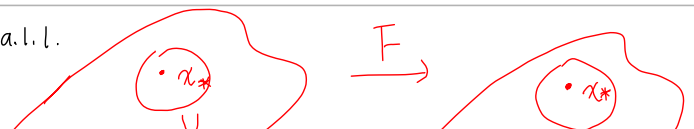
Thm 4.19 (Local contraction mapping) Let $F: \mathcal{M} \to \mathcal{M}$ be smooth map Consider the sequence defined by $x_{k+1} = F(x_k)$ $k = 0, 1, 2 \ldots$

If $x^* \in \mathcal{M}$ is a fixed pt ($F(x^*) = x^*$) and $\|DF(x^*)\| < 1$ (i.e. all singular values of $DF(x^*)$ are strictly smaller than one, then $\exists$ nbhd $\mathcal{U} \ni x^*$ s.t. if the sequence enters $\mathcal{U}$, it stays in $\mathcal{U}$ and converges to x^* at least Linearly with factor at most $\|DF(x^*)\|$.

Explicitly ① with R an arbitrary retraction on $\mathcal{M}$ we have

$$\lim_{k\to\infty} \frac{\|\xi_{k+1}\|}{\|\xi_k\|} \le \|DF(x^*)\| \quad \text{where } \xi_k \text{ is well defined by } R_{x^*}(\xi_k) = x_k \text{ for } k \text{ large enough.}$$

② If choose $R = \text{Exp}$, then we have $x_k \to x^*$ a.l.l.
since $\|\xi_k\| = \text{dist}(x^*, x_k)$



Cons: ① If $\|DF(x^*)\| = 0$, the convergence is at least quadratic

② smoothness of F can be reduce to locally smooth near x^*.

pf of Thm 4.19. choose $r > 0$ s.t R_{x^*} is a diffeo from $B(x^*, r)$ to $U = R_{x^*}(B(x^*, r))$. Consider $F^{-1}(U)$. It is an open nbhd of x^* so we can select $r' \in [0, r]$, such that $U' \triangleq R_{x^*}(B(x^*, r')) \subseteq F^{-1}(U)$

By $r' \le r \Rightarrow U' \subseteq U$.

Assume $x_k \in U'$, then $x_{k+1} = F(x_k) \in U \Rightarrow \exists!\ \xi_k, \xi_{k+1} \in T_{x^*}M$ s.t

$\xi_k = R_{x^*}^{-1}(x_k)$, $\xi_{k+1} = R_{x^*}^{-1}(x_{k+1})$. Consider $\widetilde{F}: B(x^*, r') \to B(x^*, r)$

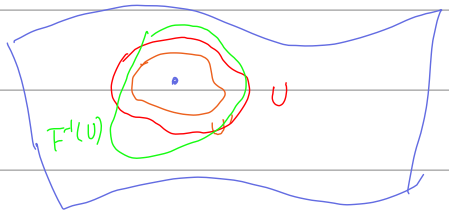
where $\widetilde{F} = R_{x^*}^{-1} \circ F \circ R_{x^*}$ $\quad T_{x^*}M \xrightarrow{R_{x^*}} M \xrightarrow{F} M \xrightarrow{R_{x^*}^{-1}} T_{x^*}M$

$B(x^*, r') \to U' \xrightarrow{F} F(U') \subseteq U \longrightarrow$ a subset of $B(x^*, r)$

$\cap$

$B(x^*, r)$





$\widetilde{F}$ satisfies the following properties:

① $\widetilde{F}: B(x^*, r') \to B(x^*, r)$ is smooth, since each is smooth.

By ①, we have Taylor expansion $\widetilde{F}(\vec{v}) = \widetilde{F}(0) + D\widetilde{F}(0)\cdot\vec{v} + E(v)$ $\forall v \in B(x^*, r')$

$\|E(v)\| = O(\|v\|^2) \le C\|v\|^2$

② $D\widetilde{F}|_0 = DR_{x^*}^{-1} \circ DF \circ DR_{x^*}|_0 = DF(x^*)$

③ $\xi_{k+1} = \widetilde{F}(\xi_k)$

$\Rightarrow \xi_{k+1} = \widetilde{F}(\xi_k) = DF(x^*)[\xi_k] + E(\xi_k)$

$\Rightarrow \|\xi_{k+1}\| \le \|DF(x^*)[\xi_k]\| + \|E(\xi_k)\| \le (\|DF(x^*)\| + C\|\xi_k\|)\|\xi_k\|$

Since $\|DF(x^*)\| < 1$, we can choose r' small enough s.t. $\|DF(x^*)\| + Cr' < 1$

$\Rightarrow \|\xi_{k+1}\| < \|\xi_k\| \Rightarrow$ if $x_k \in U'$ then $x_{k+1} \in U'$

$$\frac{\|\xi_{k+1}\|}{\|\xi_k\|} \le (\|DF(x^*)\| + C\|\xi_k\|) \le \|DF(x^*)\| + Cr' < 1$$

(C: constant)

$\Rightarrow \|\xi_k\| \to 0$ as $k \to \infty$.

$$\lim_{k\to\infty} \frac{\|\xi_{k+1}\|}{\|\xi_k\|} \le \lim_{k\to\infty} \|DF(x^*)\| + C\|\xi_k\| = \|DF(x^*)\|$$

Now assuming $\|DF(x^*)\| = 0 \Rightarrow \lim_{k\to\infty} \frac{\|\xi_{k+1}\|}{\|\xi_k\|} \le C$ which is a.l.q.

Remark: the key idea is to use local diffeo to transfer the iteration to the tangent space where standard contraction arguments apply and translates the result back to M.

Recall: Hess f ($(0,2)$ tensor version)

The Hessian of a smooth function $f: M \to \mathbb{R}$ is

$$\boxed{\text{Hess} f = \nabla df}$$ when ∇ the induced Levi-Civita on

$\nabla_X W(Y) = X(W(Y)) - W(\nabla_X$

Hess f : $\mathfrak{X}(M) \times \mathfrak{X}(M) \longrightarrow C^\infty(M)$

$(X, Y) \longmapsto \text{Hess} f(X,Y) = \nabla df(X,Y) = \nabla_X df(Y)$

Hess f $(1,1)$ tensor version. For $f: M \to \mathbb{R}$

Hess $f(x)$: $T_xM \to T_xM$ linear

$[V] \mapsto \nabla_V(\text{grad} f)$

$$\boxed{\text{Hess} f = \nabla (df)^\# = \nabla(\text{grad} f)}$$

$\text{Hess} f(x) = \nabla (df)^\# = (\nabla df)^\# \Rightarrow (\nabla df)(v,w) = g(\text{Hess} f(v), w)$

Let $\{e_1, e_2 \cdots e_n\}$ be an orthonormal basis and H is the matrix of Hess f under $\{e_1, \cdots e_n\}$

then H satisfies $H_{ij} = (\nabla_{e_i} df) e_j = \langle \text{Hess} f(e_i), e_j \rangle$

$\nabla(\nabla f)$

Thm 4.20. $\mathcal{M}$ with retraction R. Assume $x^* \in \mathcal{M}$ satisfies

① $\nabla f(x^*) = 0$ ② $\mathrm{Hess} f(x^*) \succ 0$

Let $0 < \lambda_{min} \le \lambda_{max}$ be the smallest and largest eigenvalues of $\mathrm{Hess} f(x^*)$, and $\kappa = \frac{\lambda_{max}}{\lambda_{min}}$. Set $L > \frac{1}{2}\lambda_{max}$. Given $x_0 \in \mathcal{M}$, constant step-size RGD iterates $x_{k+1} = F(x_k)$ with $F(x) = R_x\left(-\frac{1}{L}\,\mathrm{grad} f(x)\right)$. There $\exists\ U \ni x^*$ s.t. if the above sequence enters U, then it stays in U and converge to x^* at least linearly. If $L = \lambda_{max}$, the linear convergence factor is at most $1 - 1/\kappa$

pf: The key is doing differentiate of $F(x)$ $\quad F = R \circ G$

Now $\quad DF|_x = DR|_{G(x)} \circ DG|_x$.

Where $R: T\mathcal{M} \to \mathcal{M}$; $DR|_{G(x)}: T_{G(x)}T\mathcal{M} \to T_{F(x)}\mathcal{M}$

$\to G(x^*) = (x^*, 0)$

and $G: \mathcal{M} \to T\mathcal{M}$, $DG|_x : T_x\mathcal{M} \to T_{G(x)}T\mathcal{M}$

First compute $DG|_x$. Let $r(t)$ be a curve on $\mathcal{M}$ s.t $r(0) = x$ and $\dot r(0) = v \in T_x\mathcal{M}$, then $G(r(t)) = (r(t), -\frac{1}{L}\xi(t))$ where $\xi(t) \in T_{r(t)}\mathcal{M}$

Since $T\mathcal{M}$ is a vector bundle, $\Rightarrow DG|_x(\dot r(0)) = \frac{d}{dt}\Big|_{t=0} G(r(t)) = \left(\frac{d}{dt}\Big|_{t=0} r(t), -\frac{1}{L}\frac{d}{dt}\Big|_{t=0}\xi(t)\right) = \left(\dot r(0), -\frac{1}{L}\nabla_{\dot r(0)}\xi(t)\Big|_{t=0}\right) = \left(v, -\frac{1}{L}\mathrm{Hess} f(x)[v]\right)$

$\Rightarrow dF|_x(v) = dR_{Gx}\left(v, -\frac{1}{L}\mathrm{Hess} f|_x[v]\right)$ Now let $x = x^*$

$dF(x^*)[v] = dR_{(x^*,0)}\left(v, -\frac{1}{L}\mathrm{Hess} f(x^*)[v]\right)$

looking at next Lemma

$= v - \frac{1}{L}\mathrm{Hess} f(x^*)[v]$

$\Rightarrow dF(x^*) = \mathrm{Id} - \frac{1}{L}\mathrm{Hess} f(x^*)$ $\Rightarrow$ it is self-adjoint

$\|DF(x^*)\| = \max\left(\left|1 - \frac{\lambda_1}{L}\right|, \left|1 - \frac{\lambda_n}{L}\right|\right)$

Thus, F satisfies the condition in Thm 4.19

Lemma 4.21